Independent mathematical / phenomenological audit

Integer-Constrained Parameter Scans

Lepton masses, proton mass, and the Koide cone

Source: Alexander Kritov, *Clifford Geometry and Mass Relations: From Charged Leptons to the Proton*, English manuscript v13, 31 pp.
Purpose: Identify which inputs are structural integers, which are real coefficients, and how the observable predictions change when *explicitly specified assumptions* are relaxed.

Crucial rule. An integer delivered by the matrix dimension, Clifford trace, or colour representation is *not* treated as a continuously adjustable coefficient. The strict H1–H2 construction fixes 14, 17, and 9; it also fixes its real normalizations to unity. Plots labelled *diagnostic* change a physical postulate, not a mathematical identity.

Principal numerical findings

- **Strict original model:** zero tunable parameters after the empirical m_e and Thomson α are supplied. It predicts two lepton ratios and one proton ratio conditionally on its prescriptions.
- **Integer lepton weight:** restricting a diagnostic response weight $n = 1 + 16g$ to $n \in \mathbb{Z}$, with complete correction $\eta = 1$, singles out $n = 17$. The neighbouring integers miss the tau target by about 104 MeV after fitting the muon.
- **Continuous lepton diagnostics at $n = 17$:** the two ratios determine $(\lambda, \eta) = (1.000008277, 1.001399651)$ at their quoted central values. The original $(1, 1)$ is already inside their measurement tolerances.
- **Koide:** exact $Q_K = 2/3$ together with the central muon ratio gives $(\lambda, \eta) = (0.999996291, 0.999370730)$, predicting $m_\tau c^2 = 1776.9690$ MeV, only 0.43 PDG standard deviations above its central value.
- **Proton:** the integer coefficient choice $(u, q) = (1, 1)$ is close to the measured ratio; among tested integer q, it is preferred by reference-only errors, but several choices survive when the quoted uncertainty of the α^{-1} input is propagated.

Fixed numerical inputs and conventions

Quantity	Central value	Quoted standard uncertainty
α^{-1} (CODATA 2022)	137.035999177	2.1×10^{-8}
$m_e c^2$ (CODATA 2022)	0.51099895069 MeV	1.6×10^{-10} MeV
m_μ/m_e (CODATA 2022)	206.7682827	4.6×10^{-6}
$m_\tau c^2$ (PDG 2026 as quoted)	1776.93 MeV	0.09 MeV
m_p/m_e (CODATA 2022)	1836.152673426	3.2×10^{-8}

Important: CODATA recommended constants can be correlated. Combined errors in the following proton figures use an *illustrative independent-input approximation*, not the full CODATA adjustment covariance. All confidence contours are conditional on the diagnostic family and Gaussian error convention.

1. The invariant integers: no smooth tuning of 14 or 17

For the manuscript's fixed complex Clifford module, $\dim W = 16$. Its two distinguished lines leave a 14-dimensional harmonic block. With the ordinary trace and the unit graph weight in H1,

$$(d_e, d_\mu, d_\tau) = (1, 14, 1), \qquad A = P_\mu + 17P_\tau.$$

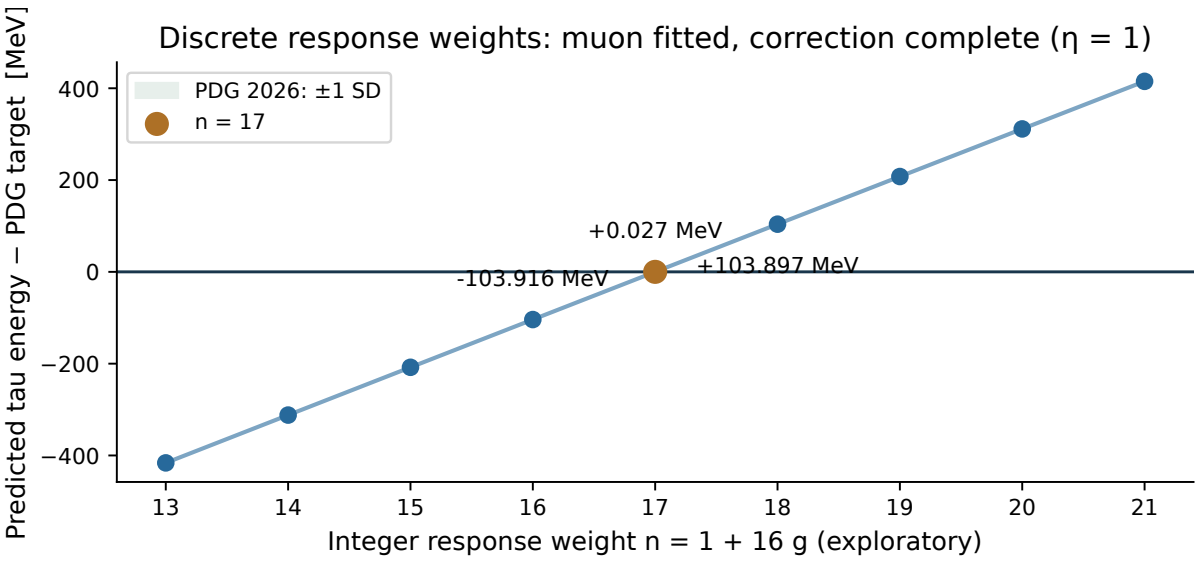
Thus 14 and $17 = 1 + 16$ are **fixed integers**, not two free real fitting parameters. Section 11.1 separately proposes a diagnostic $A_g = P_\mu + (1 + 16g)P_\tau$ with real $g > 0$. To impose an additional integrality rule on that *modified* response, set $n = 1 + 16g \in \mathbb{Z}_{\geq 2}$. This does not preserve the unit graph normalization of H1 except at $n = 17$.

The diagnostic lepton mass law becomes

$$\rho_\mu = \frac{\kappa}{\lambda} + \frac{\eta n}{14}, \qquad \rho_\tau = n\left(\frac{\kappa}{\lambda} - \eta\right), \qquad \kappa = \frac{3}{2\alpha}.$$

Holding the complete correction at $\eta = 1$ and choosing the *real* radius multiplier λ to match the muon gives the exact reduction

$$\lambda_\mu(n) = \frac{\kappa}{\rho_\mu^{\text{obs}} - n/14}, \qquad \rho_\tau(n \mid \mu, \eta = 1) = n\rho_\mu^{\text{obs}} - n\left(1 + \frac{n}{14}\right).$$



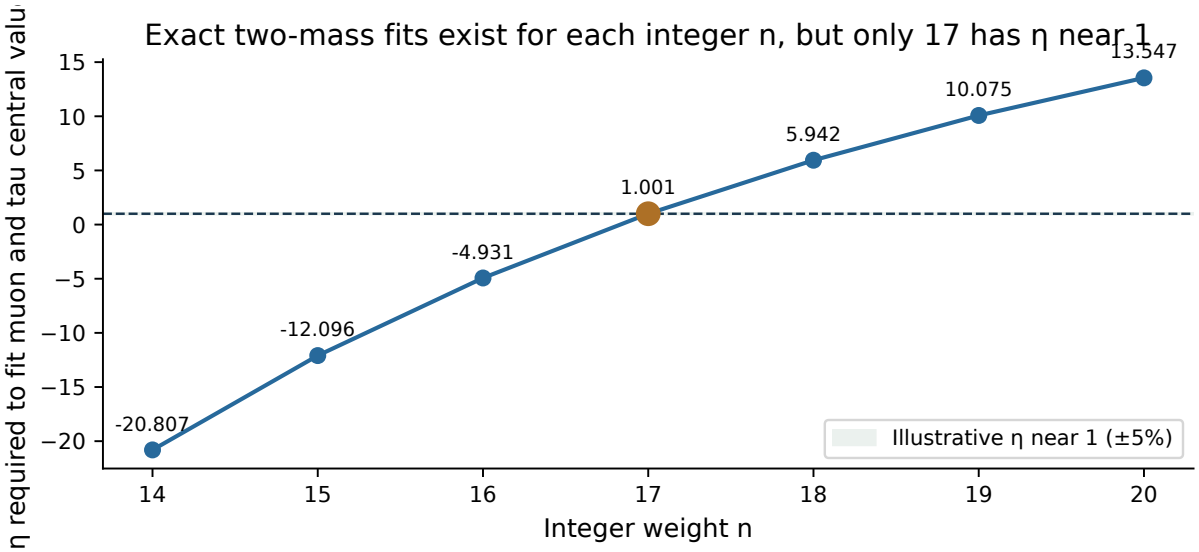
Inference. With one continuous adjustment (λ) to fit the muon, the integer alternatives $n = 16$ and 18 miss the tau by -103.916 and $+103.897$ MeV respectively. The original $n = 17$ gives $+0.026923$ MeV, well inside the 0.09-MeV PDG band. The discrete separation is genuine *within this constructed integer family.*

Status. In the strict original H1 graph norm, even the integer variation is disallowed: the only admitted weight is 17.

2. What if the correction is also allowed to vary?

With integer n, but *two* continuous diagnostic parameters λ and η, the muon and tau central targets can be fitted exactly for every admissible integer n. The inversion is elementary:

$$x_n = \frac{\rho_\tau^{\mathrm{obs}} + 14\rho_\mu^{\mathrm{obs}}}{n+14}, \qquad \lambda_n = \frac{\kappa}{x_n}, \qquad \eta_n = \frac{14}{n}\left(\rho_\mu^{\mathrm{obs}} - x_n\right).$$



Integer n	λ_n	η_n	Interpretation
16	0.967749945	-4.931288	Negative correction multiplier
17	**1.000008277**	**1.001399651**	**Near the complete-response rule**
18	1.032266609	5.941829	Correction about six times stronger

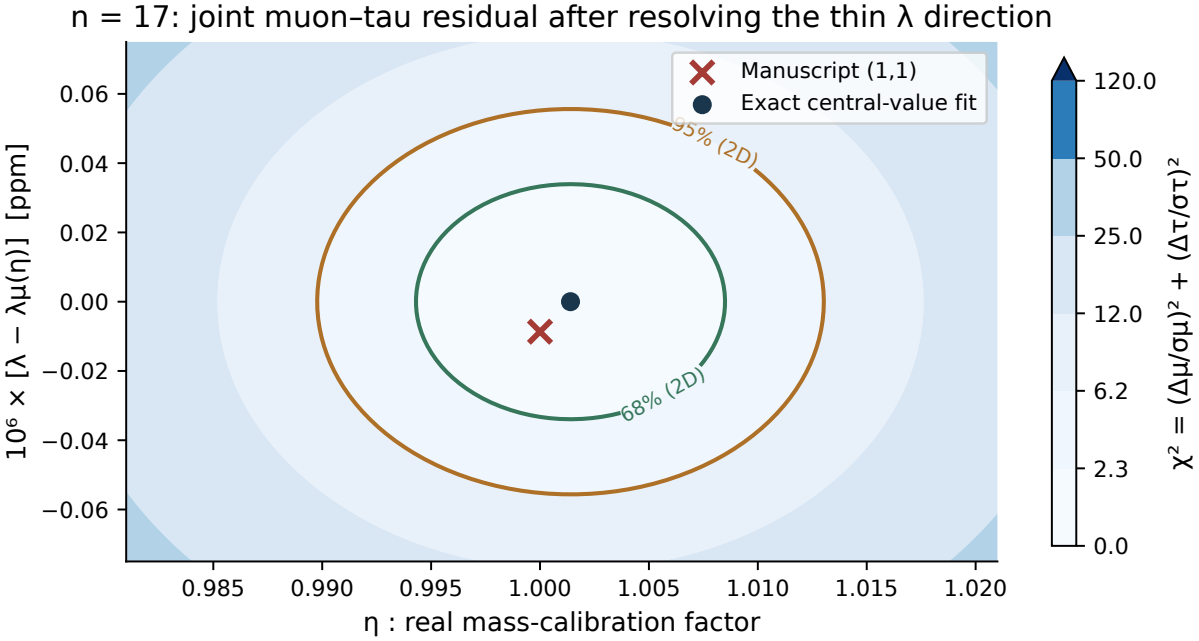
Inference. Integrality alone does not yield algebraic uniqueness if both real multipliers can be freely fitted. But $n = 17$ is exceptionally distinguished by $\lambda \simeq 1$ and $\eta \simeq 1$. In the manuscript, $\eta = 1$ is part of H2, not a prior inferred from data.

3. Local output constraint: how narrow is the lepton fit?

Now keep $n = 17$ *exactly*. The active diagnostic domain is two-dimensional, (λ, η). Let

$$\lambda_\mu(\eta) = \frac{\kappa}{\rho_\mu^{\rm obs} - 17\eta/14}, \quad \chi^2 = \left(\frac{\rho_\mu - \rho_\mu^{\rm obs}}{4.6 \times 10^{-6}}\right)^2 + \left(\frac{\rho_\tau - \rho_\tau^{\rm obs}}{0.09/m_e}\right)^2 .$$

The muon measurement makes the fit corridor almost invisible on ordinary λ scales. Consequently, the ordinate below is the displacement $10^6[\lambda - \lambda_\mu(\eta)]$, in parts per million, rather than λ itself.



The solid curves denote the conventional two-parameter Gaussian increments $\Delta\chi^2 = 2.30$ and 6.18, under the stated uncertainty model. The central exact-fit point is $(\lambda, \eta) = (1.0000082770, 1.0013996507)$. The original postulates are at $(1, 1)$; their combined residual is about $\chi^2 \simeq 0.24$, so no parameter adjustment is *required* by the reported uncertainties.

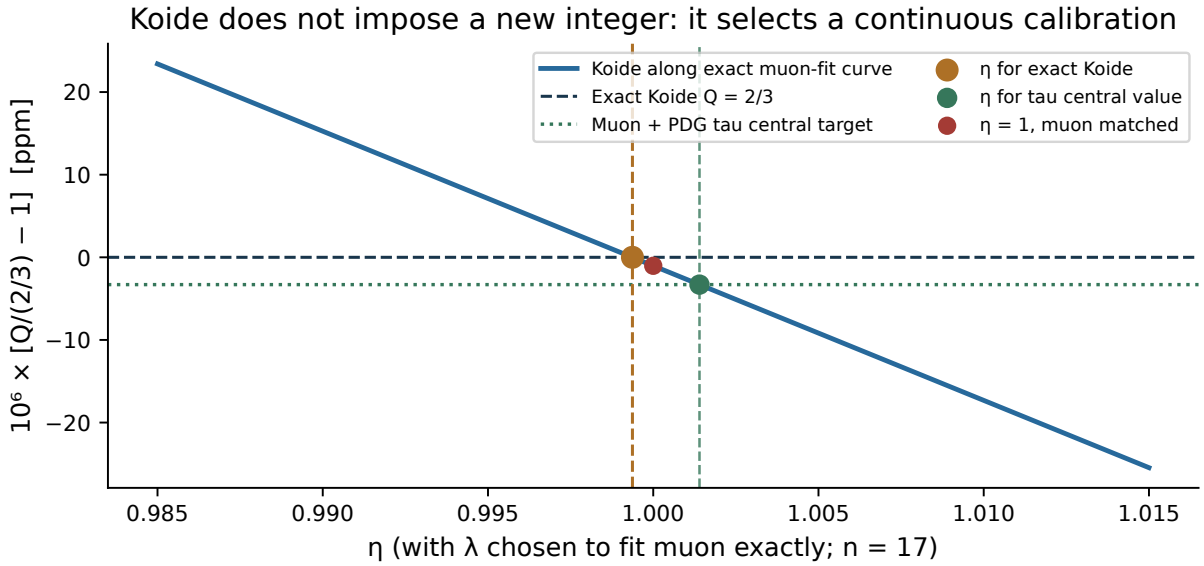
Inference. At exactly matched muon ratio, the tau ratio varies as $\rho_\tau = 17\rho_\mu^{\rm obs} - \frac{527}{14}\eta$. Its PDG uncertainty corresponds to roughly $\sigma_\eta \simeq 0.00468$ if λ is simultaneously adjusted. The two ratios strongly constrain one combination, but do not establish $\lambda = \eta = 1$ as fundamental constants.

4. Koide in the same integer-constrained space

Use the normalized Koide expression

$$Q_K = \frac{1+\rho_\mu+\rho_\tau}{(1+\sqrt{\rho_\mu}+\sqrt{\rho_\tau})^2}, \qquad \mathcal{K} = 10^6\left(\frac{Q_K}{2/3}-1\right).$$

The original integer $n = 17$, $\lambda = \eta = 1$ predicts $\mathcal{K} = -1.024877$ ppm. The selected muon and PDG tau central values give $\mathcal{K} = -3.304890$ ppm. Fix ρ_μ to its measured central value by adjusting $\lambda = \lambda_\mu(\eta)$; then Q_K is a one-variable function of the real multiplier η, without changing any integer.



Exact Koide is $\mathcal{K} = 0$. Solving its positive large-τ branch and the muon equation gives

$$\eta_K = 0.999370729841, \quad \lambda_K = 0.999996291334, \quad m_\tau c^2 = 1776.969027 \text{ MeV}.$$

The central two-mass fit instead has $\eta = 1.001399651$, $\lambda = 1.000008277$. The difference in tau mass between exact Koide and the central PDG value is 0.0390 MeV, only 0.43 PDG standard deviations. No separate integer condition forces this point: exact Koide is an *additional continuous mass constraint*. It is also not statistically independent of the two masses used to calculate it.

Inference. The cone-angle picture $Q_K = 2/3 \Leftrightarrow \theta = 45°$ is geometric, but the Clifford response postulates do not enforce that angle. The measured tau uncertainty presently accommodates exact Koide and the author's nearly exact Koide simultaneously.

5. Proton: enforce integer coefficients as a discrete test

The manuscript fixes the colour-singlet response $K_c = 9$ and a $3^3 = 27$-dimensional colour carrier. Those are *not scanned*. It also postulates a positive 2×2 return matrix

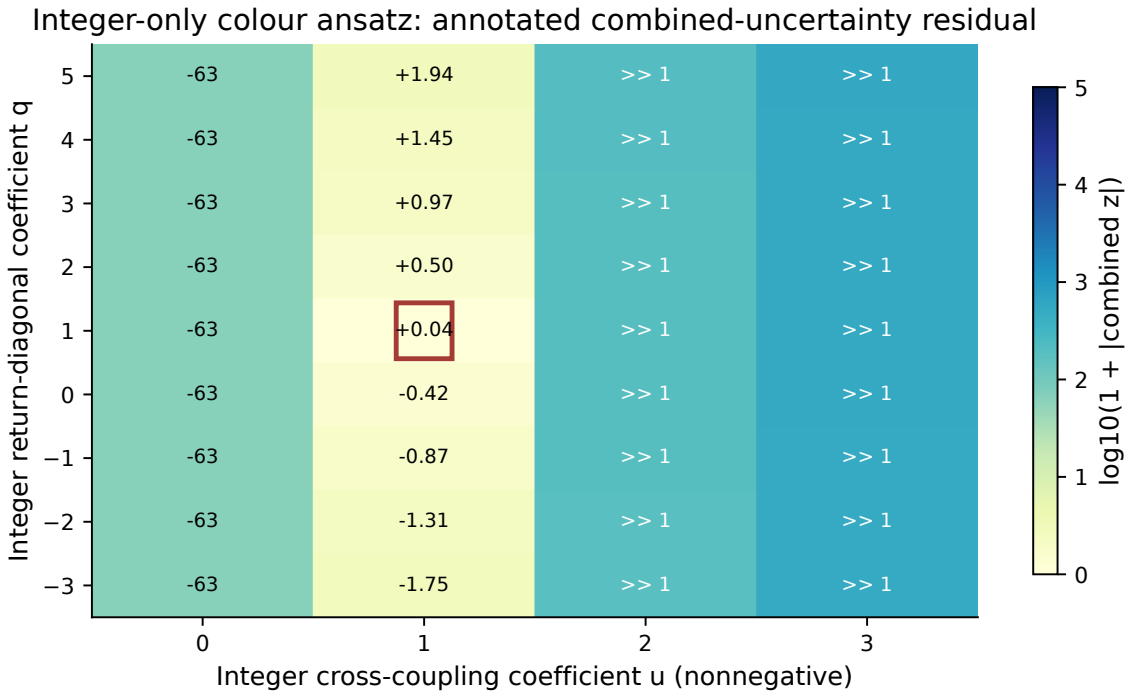
$$M_\alpha = \begin{pmatrix} 1 & -\alpha \\ -\alpha & 1-\alpha \end{pmatrix}.$$

To probe coefficient freedom while implementing a strict integer rule, consider the **counterfactual integer ansatz**

$$M_{u,q} = \begin{pmatrix} 1 & -u\alpha \\ -u\alpha & 1-q\alpha \end{pmatrix}, \qquad u \in \mathbb{Z}_{\geq 0}, \quad q \in \mathbb{Z},$$

with positivity $1 - q\alpha - u^2\alpha^2 > 0$. The colour algebra, singlet, 9 and 27 all stay fixed; the resulting conditional mass ratio is

$$\rho_p(u,q) = \frac{27}{2\alpha} - \frac{83}{6} + \frac{u^2\alpha^2}{3(1-q\alpha)}.$$



Numbers in the cells are $(\rho_p - \rho_p^{\rm obs})/\sigma_{\rm comb}$, where $\sigma_{\rm comb}$ is the illustrative combined error described next; "$\gg 1$" marks large failures. The red square is $(u,q) = (1,1)$. Integer $u = 0$ eliminates screening and misses the measured ratio by about 63 combined standard deviations; $u = 2, 3$ overshoot much more strongly.

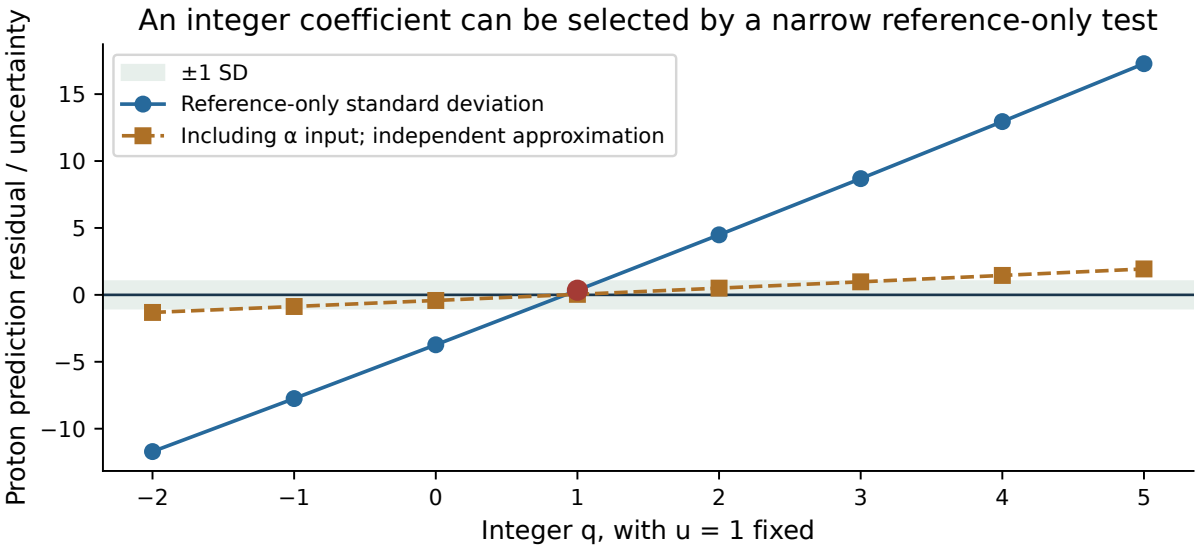
Status. The matrix coefficients are *not* representation-theoretic integer invariants. Requiring them to take integer values is an added discrete hypothesis made here at the reader's request, not a theorem of the manuscript.

6. The fine proton comparison is input-limited

At integer $u = 1$, the coefficient q changes the small screening correction by about 1.3×10^{-7} per unit. This is several times the stated CODATA reference uncertainty $\sigma_p = 3.2 \times 10^{-8}$, but *smaller* than the uncertainty propagated from the input α^{-1}. Let $x = \alpha^{-1}$. Then

$$\rho_p(1,1) = \frac{27x}{2} - \frac{83}{6} + \frac{1}{3x(x-1)}, \qquad \frac{\partial \rho_p}{\partial x} = \frac{27}{2} - \frac{2x-1}{3x^2(x-1)^2}.$$

From $\sigma_x = 2.1 \times 10^{-8}$ one obtains $\sigma_{\text{input}} \simeq 2.8350 \times 10^{-7}$. Treating input and reference errors as independent for illustration gives $\sigma_{\text{comb}} = \sqrt{\sigma_p^2 + \sigma_{\text{input}}^2} \simeq 2.8530 \times 10^{-7}$.



Integer q, $u = 1$	Reference-only z	Illustrative combined z
-1	-7.75	-0.87
0	-3.73	-0.42
1	**+0.35**	**+0.04**
2	+4.48	+0.50
3	+8.68	+0.97

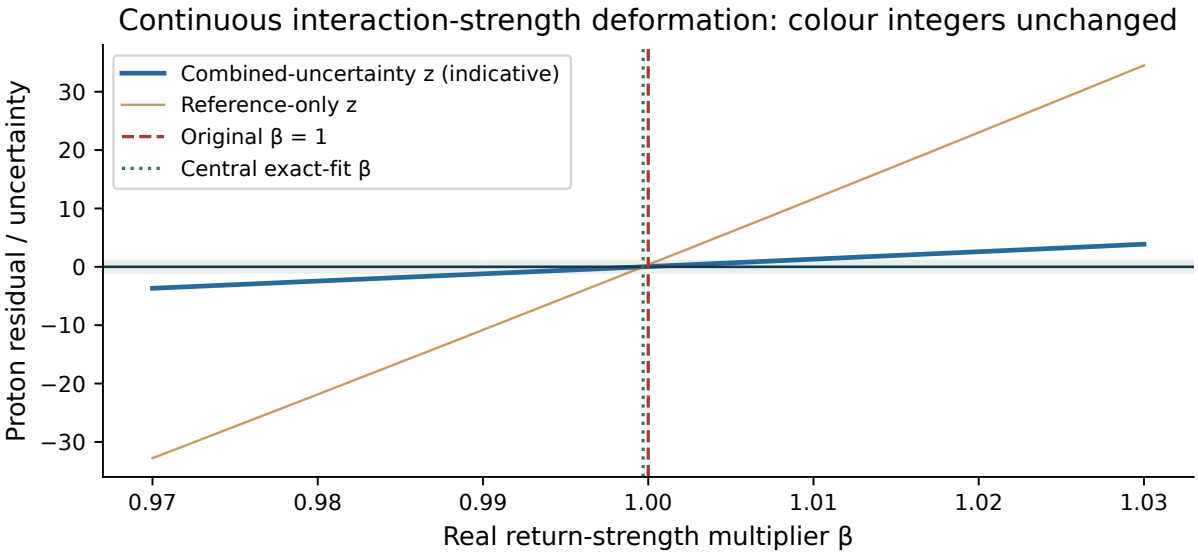
Inference. With reference-only errors the observed central match favours the integer $q = 1$. Under the illustrative propagation of the input α uncertainty, $q = -1, 0, 1, 2, 3$ all lie within one combined standard deviation. Since CODATA adjusts correlated constants, a proper joint covariance calculation is needed before assigning rigorous statistical probabilities.

7. A real coupling scan with every integer fixed

An alternative that does *not* scan integer coefficients at all is to multiply the *real* return strength α by a continuous $\beta > 0$. Retain the colour Hilbert space, $K_c = 9$, and the integer coefficients of Eq. (10.8):

$$M_{\alpha,\beta} = \begin{pmatrix} 1 & -\beta\alpha \\ -\beta\alpha & 1-\beta\alpha \end{pmatrix}, \quad \rho_p(\beta) = \frac{27}{2\alpha} - \frac{83}{6} + \frac{\beta^2\alpha^2}{3(1-\beta\alpha)}.$$

Here $\beta = 1$ recovers the exact manuscript. Changing β modifies its physical return-strength prescription; it is *not* an extra parameter of the stated theory.



The central-value fit is $\beta_* = 0.999690654$. Near $\beta \simeq 1$, the reference-only uncertainty is equivalent to $\sigma_\beta \simeq 0.000892$; illustrative propagation of the α^{-1} uncertainty increases this to approximately 0.00795. Hence the seemingly ultra-precise proton agreement does not tightly measure the underlying continuous return strength.

Comparison of constrained parameter spaces

Regime	Free parameters	Meaning of its apparent fit
Original H1–H2 + original colour ansatz	None (m_e, α are empirical inputs)	Conditional numerical agreement; no tunable residual parameters
Lepton, integer weight diagnostic	$n \in \mathbb{Z}$, $\lambda \in \mathbb{R}_+$; $\eta = 1$	Strong selection of $n = 17$ near the original geometry
Lepton, integers fixed	$\lambda, \eta \in \mathbb{R}$	A unique exact central-value fit near $(1, 1)$; Koide exact selects a different nearby point
Proton, integer coefficient ansatz	$u, q \in \mathbb{Z}$, positivity	$u = 1$ selected strongly; q poorly distinguished after input-error propagation
Proton, real strength diagnostic	$\beta \in \mathbb{R}_+$, positivity	Data constrain a continuous coupling combination, not its physical origin

Conclusion. The discrete Clifford weight 17 is robust against nearest-integer substitutions *once* a particular diagnostic prescription is chosen. The main explanatory underdetermination is elsewhere: the original choice of response, scale and mass matching, plus the independently prescribed proton return matrix. Integer-constrained plots sharpen the critique; they do not establish that these prescriptions follow from underlying dynamics.

Reproducibility and provenance. All plotted equations originate in manuscript v13, especially Eqs. (2.6), (3.3), (5.5), (6.13), (7.4), (8.3), (10.1)–(10.12), and (11.1). The integer n family and the (u, q), β colour tests are explicitly labelled counterfactual diagnostic extensions. The program `scan_figures.py` generates the figures and numerical values.